Vivek Gopalakrishnan

Massachusetts Institute of Technology
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https://vivekg.dev
Google Scholar
GitHub

Education

Harvard-MIT Program in Health Sciences and Technology — Cambridge, MA, USA
Ph.D. in Medical Engineering and Medical Physics — *2021–2026*

- Thesis: *Estimating 3D Structure and Dynamics from 2D Intraoperative Imaging*
- Advisor: Polina Golland

Johns Hopkins University — Baltimore, MD, USA
B.S. and M.S.E. in Biomedical Engineering — *2017–2021*

- Thesis: *Multiscale Statistical Hypothesis Testing for k-Sample Graph Inference*
- Advisors: Joshua Vogelstein and Carey Priebe

Employment

MIT Computer Science & Artificial Intelligence Laboratory — Cambridge, MA, USA
METEOR Postdoctoral Fellow — *2026–Present*

- Advisors: John Guttag and Adrian Dalca

Selected Publications

1. **Vivek Gopalakrishnan**, David-Dimitris Chlorogiannis, Andrew Abumoussa, Anna M. Larson, Nazim Haouchine, Darren B. Orbach, Sarah Frisken, Neel Dey, and Polina Golland. *Rapid Patient-Specific Neural Networks for X-ray to Volume Registration*. **Nature**, 2026. [paper] [project] [code] [press]
2. **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. *PolyPose: Deformable 2D/3D Registration via Polyrigid Transformations*. **NeurIPS**, 2025. [paper] [project] [code] [press]
3. **Vivek Gopalakrishnan**, Jaewon Chung, Eric Bridgeford, Benjamin Pedigo, Jesús Arroyo, Lucy Upchurch, G. Allan Johnson, Nian Wang, Youngser Park, Carey Priebe, and Joshua Vogelstein. *Multiscale Comparative Connectomics*. **Imaging Neuroscience**, 2025. [paper] [code]
4. **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. *Intraoperative 2D/3D Image Registration via Differentiable X-ray Rendering*. **CVPR**, 2024. [paper] [code]
   🏆 **Best Poster Award** at the MIT-MGB AI Cures Conference (2024).

Fellowships

CSAIL METEOR Postdoctoral Fellowship (*12 months of full funding*), MIT — 2026
HEALS Graduate Fellowship (*9 months of full funding*), MIT — 2025
Neuroimaging Training Program Grant (*12 months of full funding*), MIT — 2024
Takeda Fellowship (*9 months of full funding*), MIT — 2023
Neuroimaging Training Program Grant (*12 months of full funding*), MIT — 2022

Honors and Awards

Rising Star in AI, University of Michigan — 2025
Best Poster Award, MIT-MGB AI Cures Conference — 2024
President's Innovation Award, Society of Biomolecular Imaging and Informatics — 2023

Publications

First author on 8 papers (one co-first, marked with *) and senior author on 1. Full list on Google Scholar.

Journal and Conference Papers

1. **Vivek Gopalakrishnan**, David-Dimitris Chlorogiannis, Andrew Abumoussa, Anna M. Larson, Nazim Haouchine, Darren B. Orbach, Sarah Frisken, Neel Dey, and Polina Golland. *Rapid Patient-Specific Neural Networks for X-ray to Volume Registration.* **Nature**, 2026. [paper]

2. Janik Jehkul, Sarah Frisken, **Vivek Gopalakrishnan**, Daniel Rückert, and Nazim Haouchine. *3D Cerebrovascular Shape Completion from Biplane Angiography and CTA Prior.* **MICCAI**, 2026. [paper]

3. Federica Facente, Benjamin Billot, **Vivek Gopalakrishnan**, Manasi Kattel, Wen Wei, Polina Golland, Hervé Delingette, Nicholas Ayache, and Pierre Berthet-Rayne. *Toward Real-time Alignment of 3D CT and 2D X-ray with Multi-stage CNNs.* **Computer Assisted Surgery**, 2026. [paper]

4. **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. *PolyPose: Deformable 2D/3D Registration via Polyrigid Transformations.* **NeurIPS**, 2025. [paper]

5. **Vivek Gopalakrishnan**, Jaewon Chung, Eric Bridgeford, Benjamin Pedigo, Jesús Arroyo, Lucy Upchurch, G. Allan Johnson, Nian Wang, Youngser Park, Carey Priebe, and Joshua Vogelstein. *Multiscale Comparative Connectomics.* **Imaging Neuroscience**, 2025. [paper]

6. Sarah Frisken, **Vivek Gopalakrishnan**, David-Dimitris Chlorogiannis, Nazim Haouchine, Alexandre Cafaro, Alexandra Golby, William Wells, and Rose Du. *Spatiotemporally Constrained 3D Reconstruction from Biplanar Digital Subtraction Angiography.* **International Journal of Computer Assisted Radiology and Surgery**, 2025. [paper]

7. Charles Downs, Matthijs van der Sluijs, Sandra A.P. Cornelissen, Frank te Nijenhuis, Robert van Oostenbrugge, Wim H. van Zwam, **Vivek Gopalakrishnan**, Xucong Zhang, Ruisheng Su, and Theo van Walsum. *Improving Automatic Cerebral 3D-2D CTA-DSA Registration.* **International Journal of Computer Assisted Radiology and Surgery**, 2025. [paper]

8. **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. *Intraoperative 2D/3D Image Registration via Differentiable X-ray Rendering.* **CVPR**, 2024. [paper]
   🏆 **Best Poster Award** at the MIT-MGB AI Cures Conference (2024).

9. Joshua Marchant, Natalie Ferris, Diana Grass, Magdelena Allen, **Vivek Gopalakrishnan**, Mark Olchanyi, Devang Sehgal, Maxina Sheft, Amelia Strom, Berkin Bilgic, Brian Edlow, Elizabeth Hillman, Meher Juttukonda, Laura Lewis, Shahin Nasr, Aapo Nummenmaa, Jonathan Polimeni, Roger Tootell, Lawrence Wald, Hui Wang, Anastasia Yendiki, Susie Huang, Bruce Rosen, and Randy Gollub. *Mesoscale Brain Mapping: Bridging Scales and Modalities in Neuroimaging.* **Neuroinformatics**, 2024. [paper]

10. Sarah Frisken, Nazim Haouchine, David-Dimitris Chlorogiannis, **Vivek Gopalakrishnan**, Alexandre Cafaro, William Wells, Alexandra Golby, and Rose Du. *VESCL: An Open-Source 2D Vessel Contouring Library.* **International Journal of Computer Assisted Radiology and Surgery**, 2024. [paper]

11. Andrew Abumoussa, **Vivek Gopalakrishnan**, Benjamin Succop, Michael Galgano, Sivakumar Jaikumar, Yueh Lee, and Deb Bhowmick. *Machine Learning for Automated and Real-Time 2D-3D Registration of the Spine Using Only a Single Radiograph.* **Neurosurgical Focus**, 2023. [paper]

12. Jaewon Chung, Eric Bridgeford, Jesús Arroyo, Benjamin Pedigo, Ali Saad-Eldin, **Vivek Gopalakrishnan**, Liang Xiang, Carey Priebe, and Joshua Vogelstein. *Statistical Connectomics.* **Annual Review of Statistics and Its Application**, 2021. [paper]

13. Nian Wang, Robert Anderson, David Ashbrook, **Vivek Gopalakrishnan**, Youngser Park, Carey Priebe, Yi Qi, Rick Laoprasert, Joshua Vogelstein, Robert Williams, and G. Allan Johnson. *Variability and*

Heritability of Mouse Brain Structure: Microscopic MRI Atlases and Connectomes for Diverse Strains. **NeuroImage**, 2020. [paper]

14. **Vivek Gopalakrishnan**, Eliezer Bose, Usha Nair, Yuwei Cheng, and Musie Ghebremichael. *Pre-HAART CD4+ T-Lymphocytes as Biomarkers of Post-HAART Immune Recovery in HIV-Infected Children with or without TB Co-Infection.* **BMC Infectious Diseases**, 2020. [paper]

15. Jong Soo Lee, Elijah Paintsil, **Vivek Gopalakrishnan**, and Musie Ghebremichael. *A Comparison of Machine Learning Techniques for Classification of HIV Patients with Antiretroviral Therapy-Induced Mitochondrial Toxicity.* **BMC Medical Research Methodology**, 2019. [paper]

Peer-Reviewed Workshops

1. Noe Bertramo, Gabriel Duguey, and **Vivek Gopalakrishnan**. *DiffUS: Differentiable Ultrasound Rendering from Volumetric Imaging.* **ASMUS Workshop at MICCAI**, 2025. [paper]

2. Federica Facente, Benjamin Billot, **Vivek Gopalakrishnan**, Manasi Kattel, Wen Wei, Polina Golland, Hervé Delingette, Nicholas Ayache, and Pierre Berthet-Rayne. *Multi-stage CNN for Fast Registration of 3D Preoperative CTs to 2D Intraoperative X-rays.* **COLAS Workshop at MICCAI**, 2025. [paper]

3. Mohammadhossein Momeni*, **Vivek Gopalakrishnan***, Neel Dey, Polina Golland, and Sarah Frisken. *Differentiable Voxel-based X-ray Rendering Improves Sparse-View 3D CBCT Reconstruction.* **ML4PS Workshop at NeurIPS**, 2024. [paper]

4. **Vivek Gopalakrishnan**, Jingzhe Ma, and Zhiyong Xie. *Grad-CAMO: Learning Interpretable Single-Cell Morphological Profiles from 3D Cell Painting Images.* **CVMI Workshop at CVPR**, 2024. [paper]
   🏆 **President's Innovation Award** from the Society of Biomolecular Imaging and Informatics.

5. **Vivek Gopalakrishnan** and Polina Golland. *Fast Auto-Differentiable Digitally Reconstructed Radiographs for Solving Inverse Problems in Intraoperative Imaging.* **CLIP Workshop at MICCAI**, 2022. [paper]

Abstracts

1. Michael Hachadorian and **Vivek Gopalakrishnan**. *Recovery of 3D Component Position in Reverse Shoulder Arthroplasty from Postoperative Radiographs via 2D/3D Registration.* **American Shoulder and Elbow Surgeons Fellows' Symposium**, 2025.

Preprints

1. Victor Ion Butoi, **Vivek Gopalakrishnan**, John V. Guttag, Adrian V. Dalca, and Neel Dey. *FleXray: Universal Clinical X-ray Segmentation.* **arXiv**:2609.26756, 2026. [paper]

Invited Talks

Recovering Latent 3D Structure and Dynamics from 2D Intraoperative X-ray Images

- *MIT Medical Imaging Seminar Series*, Boston, MA 2026
- *Boston Children's Fetal-Neonatal Neuroimaging & Developmental Science Center*, Boston, MA 2026
- *Johns Hopkins University*, Baltimore, MD 2026
- *INRIA Centre at Université Côte d'Azur*, Nice, France 2026
- *Medical Physics Seminar Series*, MGB/BIDMC/Dana-Farber, Boston, MA 2026

Patient-Specific Machine Learning for 3D Intraoperative Image Guidance

- *AI for Science Symposium*, University of Michigan, Ann Arbor, MI 2025
- *United Imaging Intelligence*, Boston, MA 2025
- *Noah Medical*, San Francisco, CA 2025
- *Indian Institute of Science*, Bangalore, India 2025
- *GE HealthCare AI*, Bangalore, India 2025

Differentiable X-ray Rendering for Fast Intraoperative 2D/3D Image Registration

- *Boston Medical Imaging Workshop*, MIT, Cambridge, MA 2024
- *GE HealthCare AI*, Bangalore, India 2024
- *MIT Visual Computing Seminar*, Cambridge, MA 2024

Learning Interpretable Single-Cell Morphological Profiles from 3D Cell Painting Images

- *CVMI Workshop at CVPR*, Seattle, WA 2024
- *Society of Biomolecular Imaging and Informatics*, Boston, MA 2023

Fast Auto-Differentiable DRRs for Intraoperative Imaging Problems

- *Boston Medical Imaging Workshop*, Brigham and Women's Hospital, Boston, MA 2022
- *Image-Guided Neurosurgery Meeting*, Boston, MA 2022
- *CLIP Workshop at MICCAI*, Singapore 2022

Mentorship

- **Minsuk Choi** (M.S., University of Michigan; with Jeong Joon Park) Fall 2025–Present
  MRI-to-CT translation for radiation-free volumetric intraoperative guidance
- **Jonathan Tjandra** (Undergrad, MIT; with Polina Golland) Spring 2025–Spring 2026
  Learning to identify lung nodules in chest radiographs using synthetic data
- **Noe Bertramo** and **Gabriel Duguey** (M.S., MIT) Summer 2025
  Developing a differentiable ultrasound rendering engine [paper]
- **Zengtian Deng** (PhD, UCLA; with Bibo Shi and Tao Zhao, Noah Medical) Summer 2025
  Implementing 2D/3D registration in a deployed surgical robotic system [paper]
- **Michelle Wu** (Undergrad, MIT; with Polina Golland) Summer 2025
  A fully automated system for postoperative evaluation of shoulder replacement surgery
- **Erik Xie** (Undergrad, MIT; with Polina Golland) Fall 2024–Spring 2025
  Custom CUDA kernels for accelerated X-ray rendering
- **Hossein Momeni** (Undergrad, Truman State; with Sarah Frisken) Summer 2024–Fall 2024
  Differentiable X-ray rendering for sparse-view 3D CBCT reconstruction [paper]
  Current position: PhD student at UC Berkeley (EECS)
- Graduate Resident Advisor, McCormick Hall, MIT 2024–2026

Teaching

Advances in Computer Vision (Graduate) Spring 2025
Head Teaching Assistant, Massachusetts Institute of Technology

NeuroData Design II (Undergraduate/Graduate) Spring 2021
Head Teaching Assistant, Johns Hopkins University

NeuroData Design I (Undergraduate/Graduate) Fall 2020
Head Teaching Assistant, Johns Hopkins University

Computational Cardiology Laboratory (Undergraduate) Fall 2020
Teaching Assistant, Johns Hopkins University

Linear Algebra (Undergraduate) Fall 2020
Teaching Assistant, Johns Hopkins University

Service

PhD Admissions Committee, Harvard-MIT Health Sciences and Technology 2025
Reviewer, Winter Conference on Applications of Computer Vision (WACV) 2026
Reviewer, International Journal of Computer Vision (IJCV) 2026
Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2025–2026
Reviewer, Medical Image Analysis (MedIA) 2025–2026
Reviewer, IEEE Transactions on Medical Imaging (🏆 *Distinguished Reviewer*) 2024–2026
Reviewer, Journal of Imaging Informatics in Medicine 2024–2025
Reviewer, Information Processing in Medical Imaging (IPMI) 2025
Reviewer, Medical Image Computing and Computer Assisted Intervention (MICCAI) 2024–2025
Reviewer, Medical Imaging with Deep Learning (MIDL) 2024

Software

`xvr`: Training patient-specific 2D/3D registration models in 5 minutes 2025
Primary author and maintainer [code]
`DiffDRR`: GPU-accelerated and differentiable X-ray rendering in PyTorch 2022
Primary author and maintainer [code]

Work Experience

Noah Medical, San Carlos, CA 2025–2026
Consultant, Medical Imaging Machine Learning

- Advising the implementation of 2D/3D registration for robotic bronchoscope navigation

Xellar Biosystems, Newton, MA Summer 2023
Machine Learning Intern

- Showed that supervised deep learning models in drug discovery exploit confounding factors, and developed an interpretability metric to quantify the degree of confounding [paper]